Alejandro Soler Gonzálvez

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EDUCATION

Universidad Europea — **Madrid, Spain**
Master in Space Systems (Máster de Formación Permanente, 60 ECTS) — *2025 – 2026*

- Average grade: 8.5/10; master's thesis: 9.5/10
- Thesis: fuel-optimal guidance for autonomous rocket landing, posed as a convex program (SOCP) via lossless convexification and flown in a self-built 3-DoF simulator; reproduced a published Mars-lander benchmark (75 s, 373.76 kg of fuel)

Universidad Complutense de Madrid — **Madrid, Spain**
Bachelor of Science: Physics — *2021 – 2025*

- Bachelor's thesis (9.2/10): multi-agent Bayesian search with Gaussian-process (Kriging) regression; selected for Docta Complutense, the university's distinguished-thesis repository
- Relevant coursework: mathematical methods (ODEs, PDEs, complex analysis), multivariable calculus, linear algebra, statistical physics, computational physics (Monte Carlo, Python), statistics and data analysis; university project in quantum computing (Qiskit)

WORK EXPERIENCE

Independent Quantitative Research — **Remote**
Cross-Venue BTC Oracle Basis (Polymarket / Kalshi), published on SSRN — *Feb 2026 – May 2026*

- Identified a systematic, signed mispricing between two hourly Bitcoin prediction markets that settle on different oracles (Binance USDT vs CF Benchmarks USD), driven by the USDT/USD stablecoin basis
- Validated on 1,350 real settlements: an offset-aware entry rule removed the double-loss tail (4.6% to 0%) and isolated the $\sim$21% regime where both legs win
- Designed the 24/7 data-capture system on AWS, recording live order books from both venues

PLD Space — **Elche, Spain**
Flight Dynamics Intern — *Mar 2025 – Apr 2025 & Jun 2025 – Aug 2025*

- Built MATLAB tooling that automated recurring flight-dynamics analyses for the team
- Reworked the numerical surface-discretisation scheme behind the vehicle's aerodynamic force integration, improving its accuracy

Bolbee (incubated by MyBookUp) — **Spain**
Co-founder and Sole Developer — *Oct 2025 – Aug 2026*

- Built an AI marketing platform end-to-end (Next.js, Supabase, Stripe, multi-model AI pipeline), funded by MyBookUp under contract
- Reached a private beta with 20 active users; chose not to launch when the funding ended

EXTRACURRICULAR ACTIVITIES

NASA Space Apps Challenge — **Madrid, Spain**
Team Lead — *Oct 2025*

- Led a 3-person team that built a neural-network exoplanet classifier on NASA telescope data in 48 hours, achieving $\sim$88% classification accuracy; prepared and standardised the input data

ADDITIONAL INFORMATION

Languages:	Spanish (Native), English (C1, Cambridge CAE 186), Catalan (C1), French (A2)
Technical Skills:	Python (NumPy, pandas), MATLAB, Git, AWS; convex optimisation, Monte Carlo, Gaussian processes
Awards:	Merit scholarship (full first-year tuition; ranked 1st in high-school class); EBAU 9.74/10
Community:	Private tutor for students preparing for the university entrance exam; volunteer in the 2024 Valencia flood relief
Interests:	Sprint triathlon training, composing and producing music (piano piece performed at a public concert in Alicante), Texas Hold'em poker (regular games and occasional tournaments)